

Derbyshire Shared Care Pathology Guidelines

Diagnosis and treatment of pancreatic exocrine insufficiency in patients with known diagnosis of diabetes

- Pancreatic exocrine insufficiency (PEI) is characterised by deficiency of three major pancreatic enzymes (amylase, protease, lipase) and results in impaired digestion, nutrient malabsorption and malnutrition.
- Gastrointestinal symptoms in Diabetes are commonly attributed to the side-effects of medications, particularly metformin, glucagon-like peptide-1 (GLP-1) receptor agonists, dipeptidyl peptidase-4 (DPP-4) inhibitors (list).
- However, some degree of PEI is common in diabetes (see reference 1) and is under-investigated. If symptoms persist despite stopping or reducing the dose of possible culprit medications, an alternative cause for these symptoms should be sought.

Purpose of guideline

Investigation of altered bowel habit in people with diabetes as higher risk of coeliac/ pancreatic insufficiency

- See page 2 for flowchart outlining appropriate investigation
- Patients with diabetes and PEI should be considered for referral to diabetes clinic for assessment, particularly for the following circumstances
 - Sudden rise in HbA1c
 - On insulin
 - Not meeting HbA1c targets

Clinical features suggestive of PEI

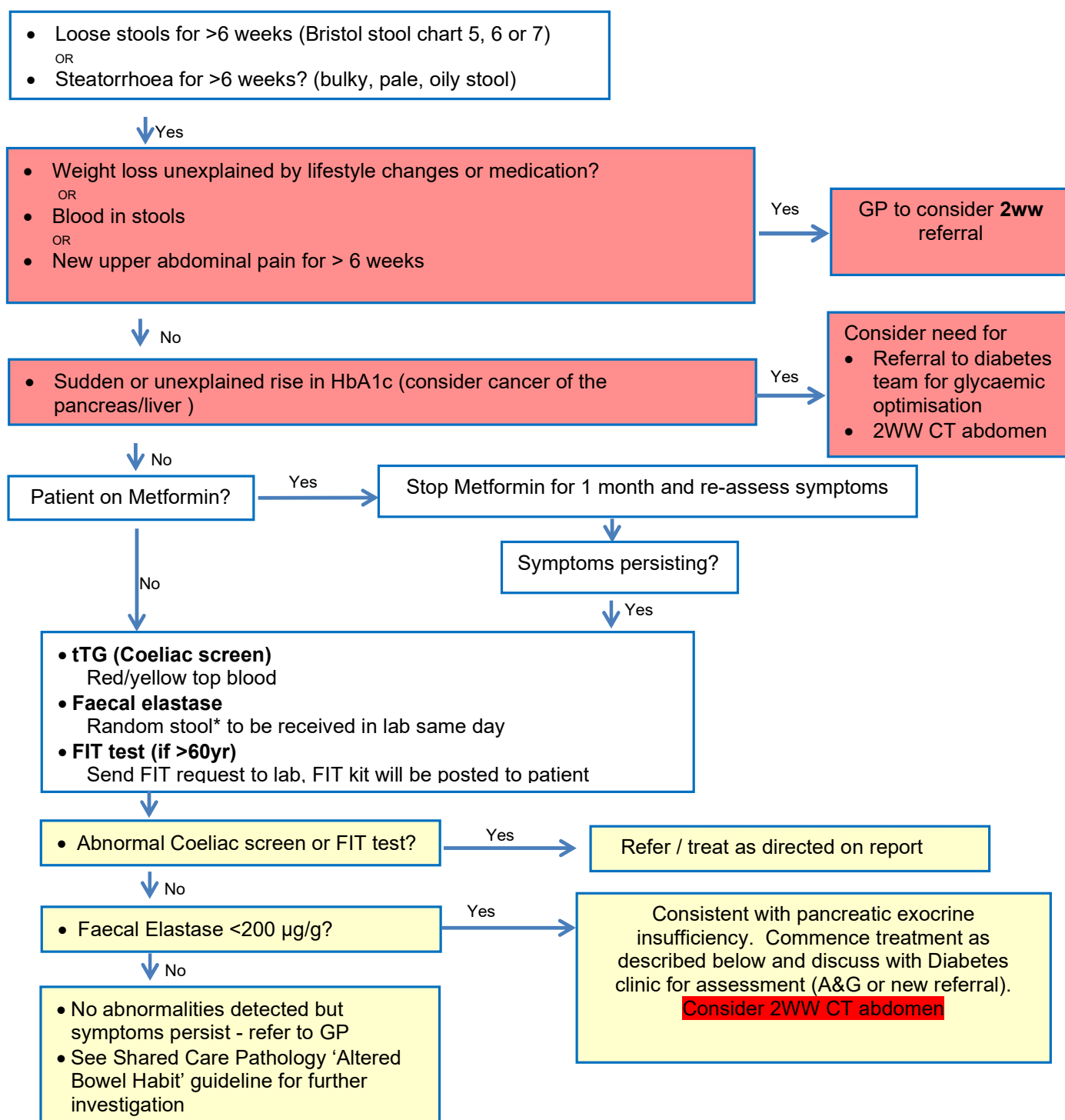
- Loose stools and scores of 5, 6 or 7 on the Bristol Stool Chart are consistent with PEI
- Steatorrhoea and weight loss typically occur only in severe cases or late-stage disease and their absence does NOT exclude a diagnosis of PEI

Rectal bleeding, unexplained weight loss and blood in the stool are not due to PEI and should be considered red flag signs warranting specialist referral

Faecal Elastase

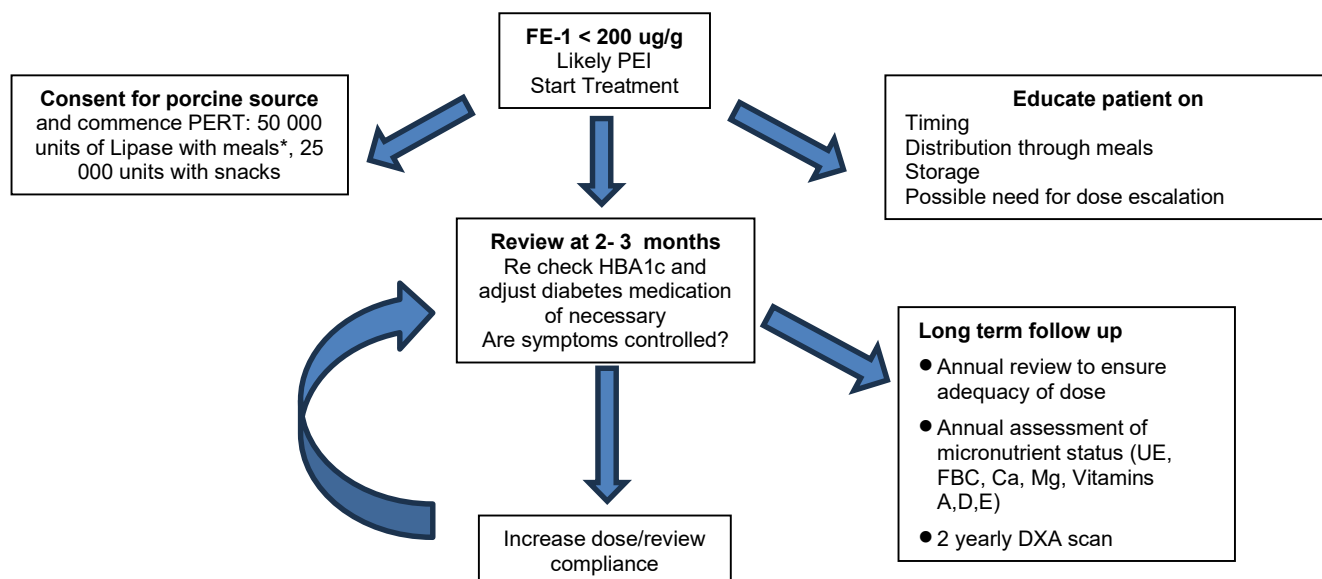
- Faecal elastase <200 µg/g is suggestive of PEI, with values <100 µg/g indicating more severe disease
- Pancreatic Enzyme Replacement Therapy (PERT) is appropriate in these patients
- Note that measurement of faecal elastase can be unreliable if the patient has very loose stools

Investigation of altered bowel habit in patients with known diabetes



*Instructions for collection of a stool sample can be found at UHDB patient information sheets [Here](#). Note watery samples can be a cause of falsely low Faecal elastase results, therefore patients should be encouraged to collect as formed a sample as possible and avoid contamination with water.

Pancreatic enzyme replacement therapy (PERT)



Notes:

- Starting dose is 1 Creon (25,000 units) with every snack or milky drink and 2-3 Creon (50,000 to 75,000 units) with every main meal
 - *a higher dose may be needed with very fatty meals such as fried foods, takeaways, cheese and pastry)
- Take with or just before food
- Strongly consider also prescribing PPI or Famotidine daily to increase efficacy at lower doses
- Patient can titrate up by 1 capsule Creon 25,000 per day with larger meals to maximum 4 caps with largest meal to achieve resolution of diarrhea if no improvement in symptoms after 2 weeks
- Note PERT is pork based. Religious/Cultural implications should be considered and patients informed for consent prior to commencing treatment
- Fibrosing colonopathy is a rare side effect of PERT. Counsel patients to seek help if they experience any of the below after commencing PERT
 - Unusual or severe
 - Stomach pain
 - Bloating
 - Nausea
 - Vomiting
 - Diarrhoea

PERT shortage: Local guidelines for prescribing PERT can be found in the JAPC guideline below
[https://www.derbyshiremedicinesmanagement.nhs.uk/assets/Clinical_Guidelines/Formulary_by_BNF_chapter_prescribing_guidelines/BNF_chapter_1/DDICB%20guide%20\(PERT\).pdf](https://www.derbyshiremedicinesmanagement.nhs.uk/assets/Clinical_Guidelines/Formulary_by_BNF_chapter_prescribing_guidelines/BNF_chapter_1/DDICB%20guide%20(PERT).pdf)

2 - 3 month review

- PERT can alter glycaemic control – recheck HbA1c and adjust diabetic medication if necessary
- If symptoms not controlled, consider increasing dose of PERT and/or refer to specialist.
- **If dose is > 100 000 units** per meal Consider changing PERT product and See Shared Care Pathology 'Altered Bowel Habit' guideline for further investigation.
- If symptoms progress or fail to respond, refer to a specialist

References:

1. Hardt P, Ewald N (2011) Experimental Diabetes Res 2011: 1–7
<https://www.pcdsociety.org/resources/details/pancreatic-exocrine-insufficiency-in-type-1-and-type-2-diabetes-more-common-than-you-think>
2. Radlinger B, Ramoser G & Kauer S Current Diabetes Reports (2020) 20: 18 [Exocrine Pancreatic Insufficiency in Type 1 and Type 2 Diabetes](#)
3. Whitcomb DC, Buchner AM and Forsmark CE. Gastroenterology 2023;165:1292–1301
[AGA Clinical Practice Update on the Epidemiology, Evaluation, and Management of Exocrine Pancreatic Insufficiency: Expert Review](#)
4. Philips ME, et al BMJ Open Gastro 2021;8:e000643
<https://bspghan.org.uk/wp-content/uploads/2023/01/consensus-for-the-management-of-pancreatic-exocrine-insufficiency-UK-practical-guidelines.pdf>

Document controls

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Secondary Care UHDB	16.01.26	Dr T Crabtree	Consultant Diabetologists
	16.01.26	Dr E Robinson	
Primary Care	15.01.26	Dr K Owen	GP and GP Clinical lead for diabetes, Derby and Derbyshire ICB
	30.01.26	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	23.01.26	Ms S Knowles	Consultant Clinical Scientist